SVM

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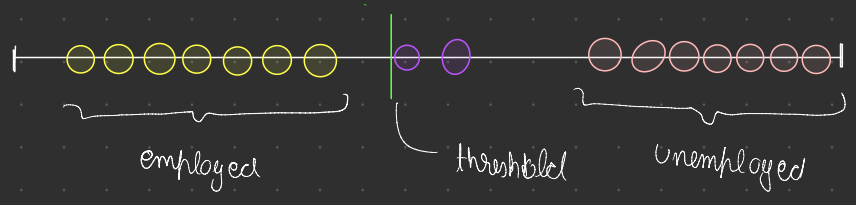
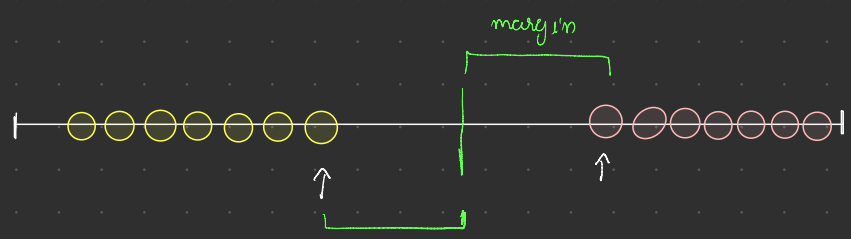
Support vector machines

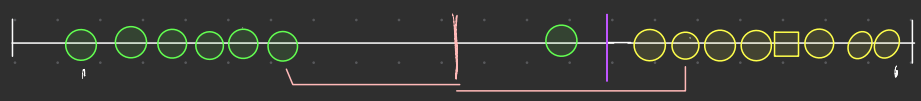




The shortest distance between the observation and the threshold is known as margin.

and we also call it as $\boxed{\text{maximum margin classifier}}$

↳ It is very sensitive to outliers



To improve this situation we used misclassification

When we allowed the misclassification and did the bias variance tradeoff, after this we will achieve the better threshold

$\boxed{\text{Support vector classifier}}$

We use SVM here

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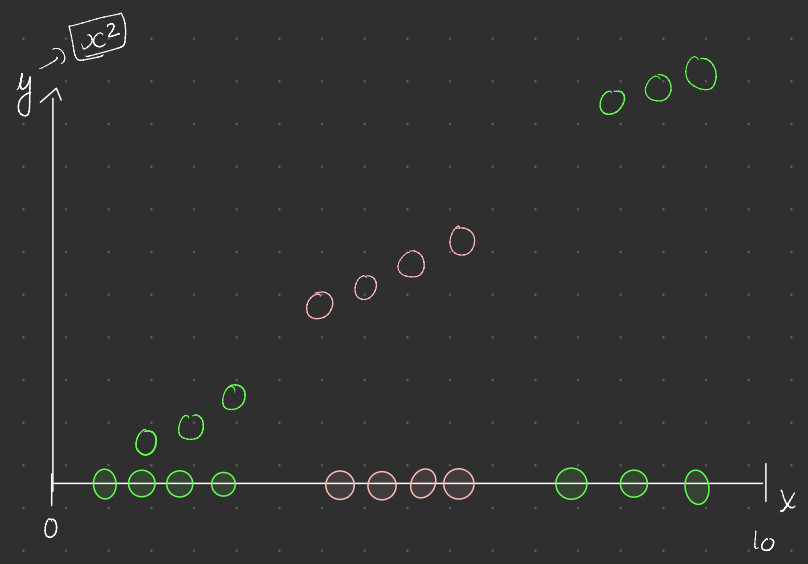

1) Start with low dimensions, 2) move the data to higher dimensions

3) Find the support vector classifier

whenever we want to change the dimension we have to use $\boxed{\text{Kernels}}$

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